Master Team Project Summer 2024
ClassCompanion
Milestone 4

Team 1

Yagubkhan Yagubkhanov, Team Lead / Github Master
yagubkhan.yagubkhanov@informatik.hs-fulda.de

Samman Adhikari, Frontend Lead

Shantanu Patil, Backend Lead

Aralikere Yogananda Gunarachana

Mariha Asif

Amber Asad

Date: 08.07.2024

Revision Summary:

Date	Revision	Summary
08.07.2024	1.0	Initial draft created

Content and structure for Milestone 4

1. Product summary
2. Usability test plan
3. QA test plan
4. Code review
5. Self-check on best practices for security
6. Self-check: Adherence to original Non-functional specs – performed by team leads

1. Product Summary

Product Name

ClassCompanion

Product Overview

ClassCompanion is a comprehensive platform designed to facilitate seamless interaction between students and tutors. By integrating essential features such as user registration, profile management, search functionality, and real-time communication, ClassCompanion aims to enhance the educational experience by providing an accessible and efficient tutoring service.

Major Committed Functions

User Registration and Login

Students and tutors must register using their university email address, with email validation to ensure eligibility, and securely log in and log out with the option to register as a student or tutor via a checkbox.

Profile Management

Users (students and tutors) can manage their profiles, update personal information, add skills, and upload profile pictures, with students restricted from searching for other students based on skills or subjects.

Tutor Search Functionality

Students can search for tutors based on subjects, keywords, experience, and rating with search results displaying a list of relevant tutors based on the search criteria.

Live Chat Feature

Students can initiate live chat sessions with tutors, with chat functionality supporting text messages, file uploads (such as documents or images), and other multimedia content, and offline messages delivered to tutors' inboxes for later viewing.

Inbox Management

Both tutors and students have access to their respective inboxes, displaying messages received from other users, including chat messages and notifications, with users able to mark messages as read, reply, and delete messages as needed.

Admin Panel

An admin panel is available for moderators to manage user accounts, monitor chat activity, handle reported issues, view user profiles, moderate content, and enforce site policies, with the ability to hide or delete inappropriate content (Unique Selling Proposition).

Unique Feature

Group Chat Functionality

Facilitates collaborative learning by enabling tutors to engage with multiple students at once, promoting group study sessions and discussions.

URL to Access the Product:

http://35.158.64.195/login

2. Usability Test Plan for ClassCompanion

This document describes a test plan for conducting a usability test during the development of ClassCompanion.

Selected Function for Testing

Chat Functionality

Test Objectives

The objective of this usability test is to evaluate the chat functionality of ClassCompanion. This includes assessing the ease of initiating a chat, sending messages, and maintaining a conversation between students and tutors. The test aims to measure user satisfaction, identify any usability issues, and ensure that the chat feature meets user expectations for an educational platform.

Test Background and Setup

System Setup: The test will be conducted using the deployed on our server version of ClassCompanion on a standard laptop or desktop computer or mobile phone with internet access. The URL for accessing the system will be provided to the participants.

Starting Point: Participants will begin at the ClassCompanion homepage, where they will log in using pre-registered accounts. They will then navigate to the chat feature.

Intended Users: The intended users for this test are students and tutors. The participants selected for this test include students pursuing different levels of higher education and a teaching assistant to ensure a diverse range of feedback.

URL of the System to be Tested: http://35.158.64.195/login (this URL will be provided to the participants prior to testing).

What is to be Measured: For this test, the primary focus will be on user satisfaction with the chat functionality. This will be evaluated through a Likert scale questionnaire, capturing user feedback on various aspects of the chat experience, including ease of use, clarity, and responsiveness.

Usability Task Description

Participants will be given the following instructions for the usability test:

1. **Task 1:** Imagine you need help with a specific academic topic (e.g., calculus homework, research paper, or lecture preparation). Initiate a chat with a tutor on ClassCompanion.
2. **Task 2:** Send a message to the tutor describing your problem or topic of interest.
3. **Task 3:** Engage in a conversation with the tutor for a few exchanges, noting any difficulties or issues encountered.

Effectiveness Measurement: Effectiveness will be measured by the participants' ability to successfully initiate a chat, send messages, and maintain a conversation without encountering major issues.

Efficiency Measurement: Efficiency will be measured by the time taken to complete each task and the number of steps required to perform each action within the chat functionality.

Likert Scale Questions for User Satisfaction:

Participants will use a Likert scale to assess their satisfaction with the chat functionality in ClassCompanion after completing the usability tasks. They will rate their experience based on the following criteria:

1. Overall Experience: How would you rate your overall experience with the chat functionality in ClassCompanion?
    - 1 (Very Poor)
    - 2 (Poor)
    - 3 (Neutral)
    - 4 (Good)
    - 5 (Very Good)
2. Interface Usability: How simple and clear was the chat interface to use?

    - 1 (Very Complicated)
    - 2 (Complicated)
    - 3 (Neutral)
    - 4 (Simple)
    - 5 (Very Simple)
3. Responsiveness: How responsive was the chat functionality during your interaction?
    - 1 (Very Unresponsive)
    - 2 (Unresponsive)
    - 3 (Neutral)
    - 4 (Responsive)
    - 5 (Very Responsive)

These questions are designed to gather insights into user satisfaction regarding the usability, clarity, and responsiveness of the chat feature in ClassCompanion.

Test Participant Details

Test Participant 1

Screening & Pre-test Questions:

- **Q1:** How old are you? - **A1:** 22
- **Q2:** What is the highest level of education you've completed? - **A2:** Bachelor's Degree
- **Q3:** What is your current occupation? - **A3:** Student (Pursuing Master's)
- **Q4:** On a scale of 1 to 5, how would you rate your level of confidence in using your laptop/PC for online communication? - **A4:** 5
- **Q5:** When was the last time you used an online chat application for educational purposes? - **A5:** Last month

Likert Scale Questions Responses:

Criteria	Response
Criteria 1	5/5

Criteria 2	5/5
Criteria 3	4/5

Post Test Questions:

- **Q1:** How was your overall experience when using the chat functionality? - **A1:** Good
- **Q2:** How simple and clear was the chat interface? - **A2:** It is very simple and clear with instructions.
- **Q3:** Did you encounter any issues during the chat? - **A3:** No, it worked without any issues.
- **Q4:** How was your experience with the responsiveness of the chat? - **A4:** Simple and clear.
- **Q5:** How was your overall experience using ClassCompanion? - **A5:** It was very easy to use, and I faced no issues during the chat process.

Test Participant 2

Screening & Pre-test Questions:

- **Q1:** How old are you? - **A1:** 24
- **Q2:** What is the highest level of education you've completed? - **A2:** Master's Degree
- **Q3:** What is your current occupation? - **A3:** Research Assistant
- **Q4:** On a scale of 1 to 5, how would you rate your level of confidence in using your laptop/PC for online communication? - **A4:** 4
- **Q5:** When was the last time you used an online chat application for educational purposes? - **A5:** Two months ago

Likert Scale Questions Responses:

Criteria	**Response**
Criteria 1	5/5
Criteria 2	5/5

Criteria 3	5/5

Post Test Questions:

- **Q1:** How was your overall experience when using the chat functionality? - **A1:** Very Good
- **Q2:** How simple and clear was the chat interface? - **A2:** It is straightforward and easy to use.
- **Q3:** Did you encounter any issues during the chat? - **A3:** No, it was smooth.
- **Q4:** How was your experience with the responsiveness of the chat? - **A4:** Very responsive.
- **Q5:** How was your overall experience using ClassCompanion? - **A5:** It was user-friendly and efficient for my needs.

Test Participant 3

Screening & Pre-test Questions:

- **Q1:** How old are you? - **A1:** 26
- **Q2:** What is the highest level of education you've completed? - **A2:** PhD Candidate
- **Q3:** What is your current occupation? - **A3:** Teaching Assistant
- **Q4:** On a scale of 1 to 5, how would you rate your level of confidence in using your laptop/PC for online communication? - **A4:** 5
- **Q5:** When was the last time you used an online chat application for educational purposes? - **A5:** Last week

Likert Scale Questions Responses:

Criteria	**Response**
Criteria 1	5/5
Criteria 2	5/5
Criteria 3	5/5

Post Test Questions:

- **Q1:** How was your overall experience when using the chat functionality? - **A1:** Excellent
- **Q2:** How simple and clear was the chat interface? - **A2:** It is very intuitive and user-friendly.
- **Q3:** Did you encounter any issues during the chat? - **A3:** No, it worked flawlessly.
- **Q4:** How was your experience with the responsiveness of the chat? - **A4:** Very quick and reliable.
- **Q5:** How was your overall experience using ClassCompanion? - **A5:** It was seamless and highly effective for my purposes.

3. QA Test Plan:

Selected Function for Testing:

Chat Functionality

Test Objectives

The objective of this QA test plan is to verify the functionality, usability, and performance of the chat feature in ClassCompanion. This includes testing message sending, receiving, real-time updates, and overall reliability across different browsers.

HW and SW Setup

- **Hardware Setup:** Standard laptops/desktops with internet connectivity.
- **Software Setup:** Latest versions of Chrome and Firefox browsers.
- **URL:** http://35.158.64.195/login (to be provided to QA testers).

QA Test Plan

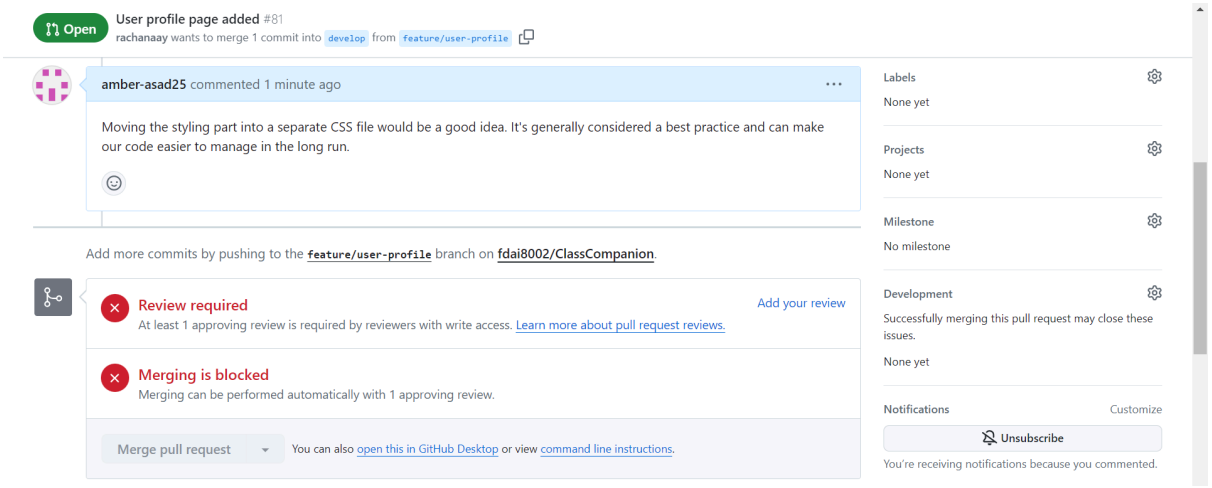
Test #	Test Title	Test Description	Test Input	Expected Correct Output	Test Results (Pass/Fail)
1	Send Message	Test sending a message from student to tutor.	Type message in chat input field	Message is sent and received by tutor	PASS (Chrome), PASS (Firefox)

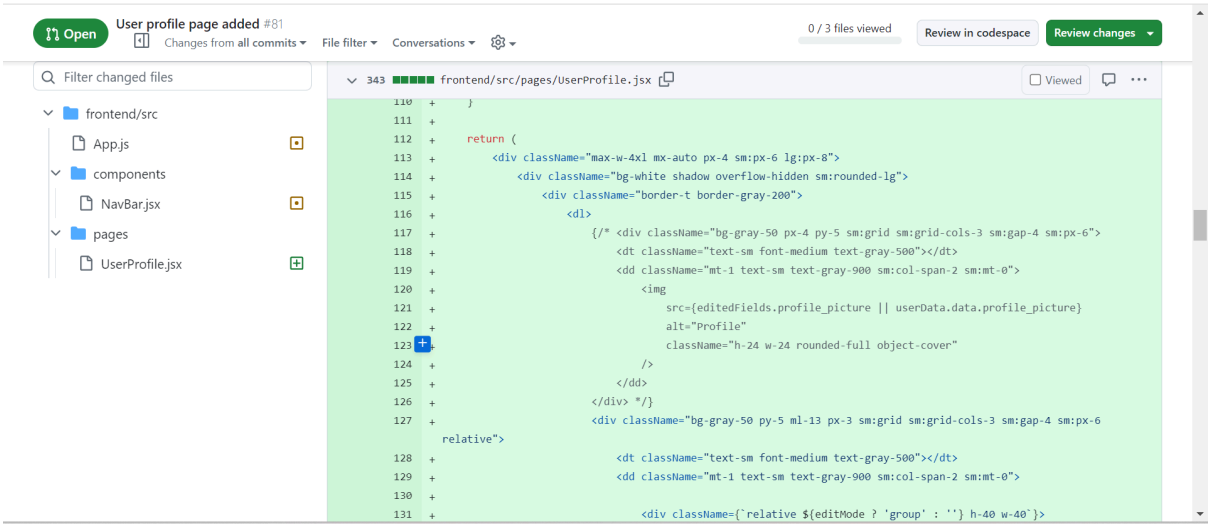
2	Receive Message	Test receiving a message from tutor to student.	Tutor sends message to student	Message is received promptly by student	PASS (Chrome), PASS (Firefox)
3	Real-time Updates	Test real-time updates when both users are online.	Send messages simultaneously	Messages appear instantly for both users	PASS (Chrome), PASS (Firefox)
4	Offline Message Delivery	Test message delivery when the recipient is offline.	Send message to offline tutor	Message is delivered upon tutor login	PASS (Chrome), PASS (Firefox)

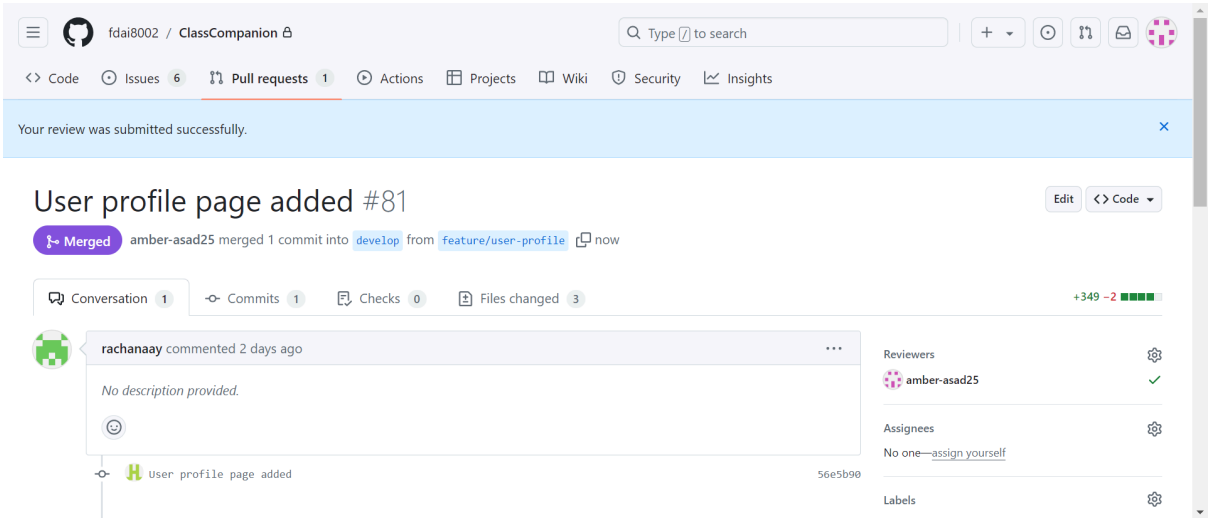
4. Code Review:

Amber Asad code review:

Aralikere Yogananda Gunarachana code reviewed by Amber Asad (https://github.com/fdai8002/ClassCcompanion/pull/81):

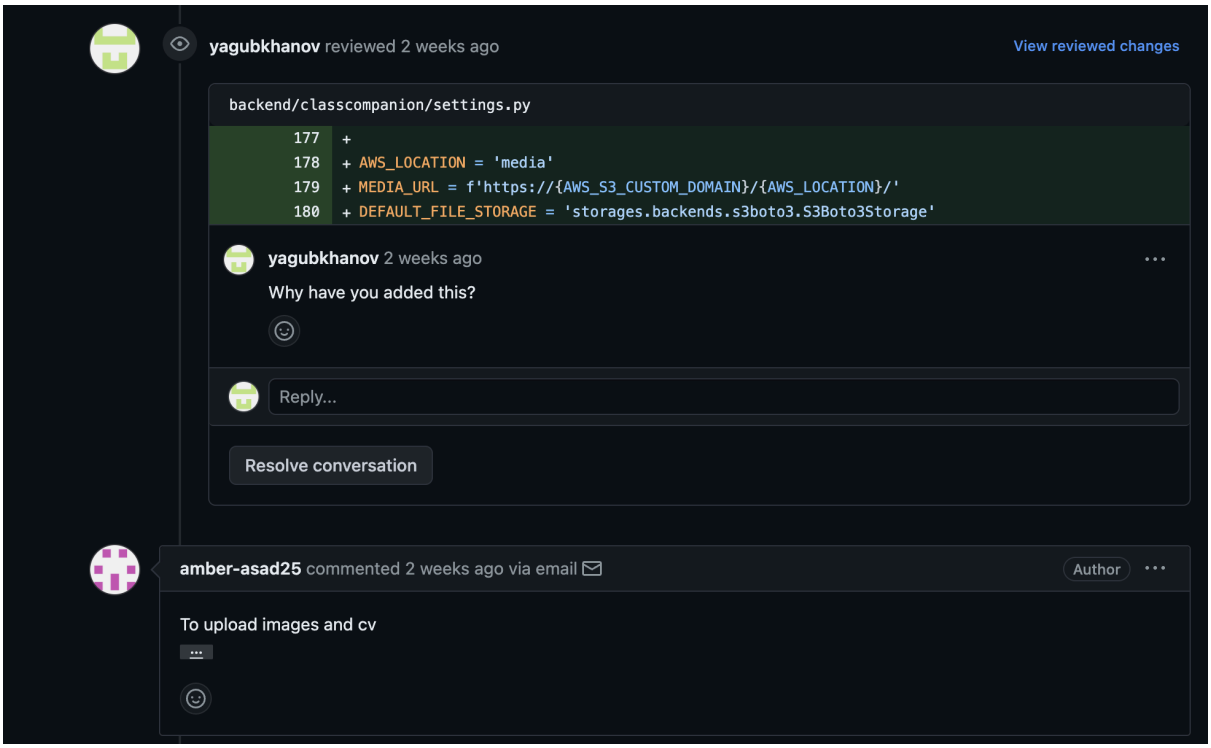




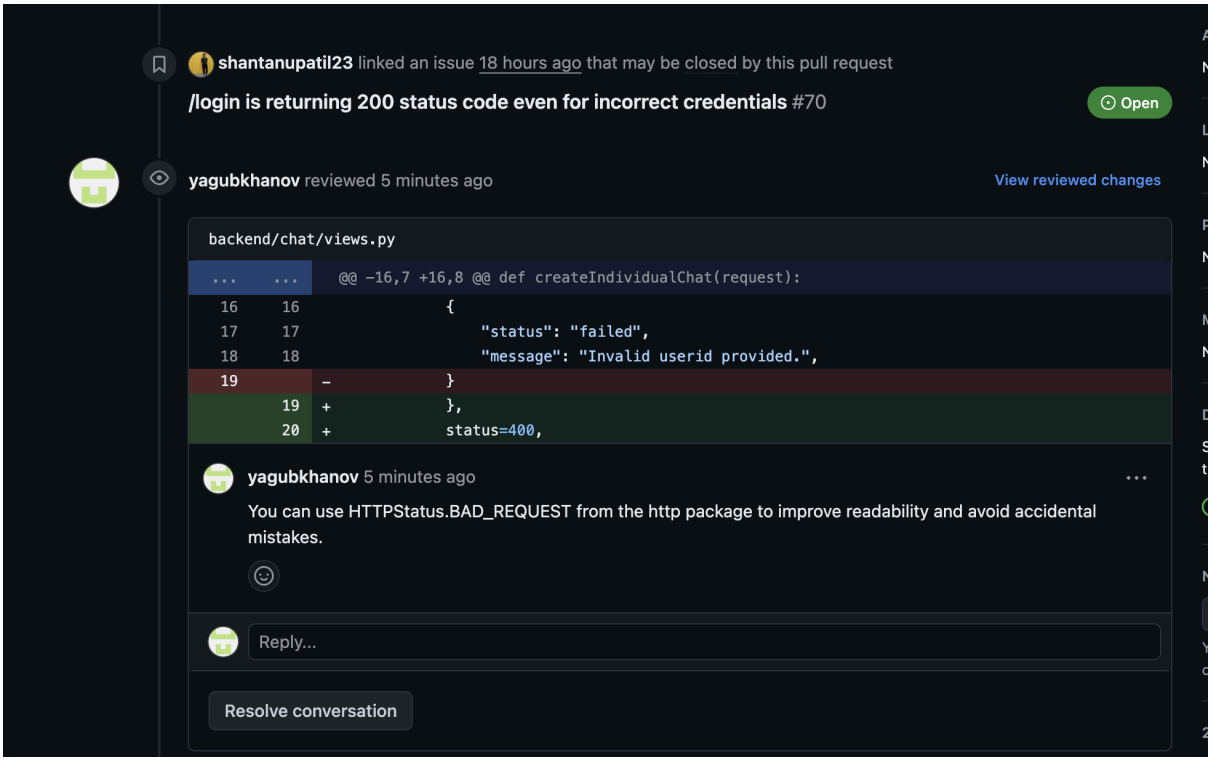


Yagubkhan Yagubkhanov code review:

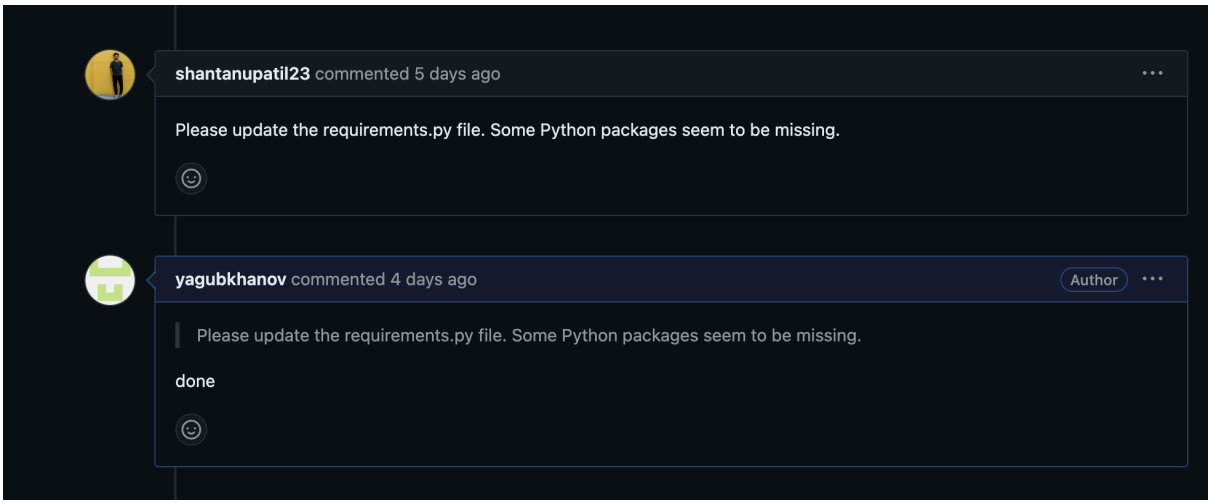
https://github.com/fdai8002/ClassCompanion/pull/45:



https://github.com/fdai8002/ClassCompanion/pull/78:

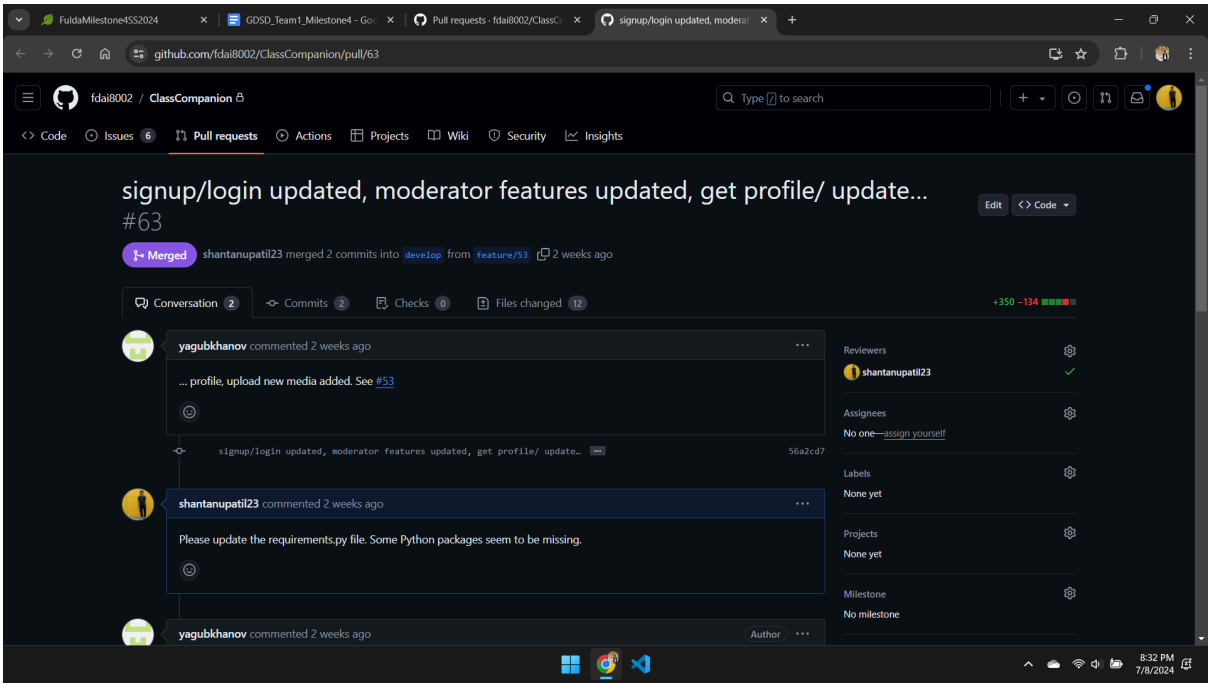


https://github.com/fdai8002/ClassCompanion/pull/63:

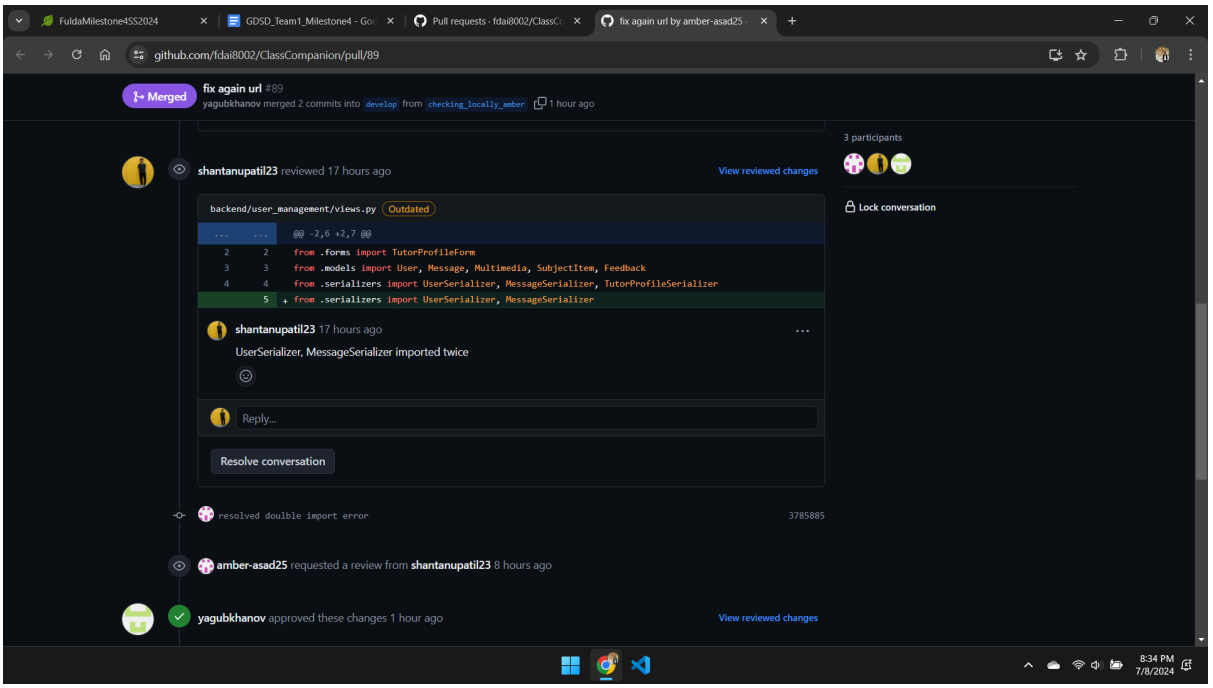


Shantanu Patil code review:

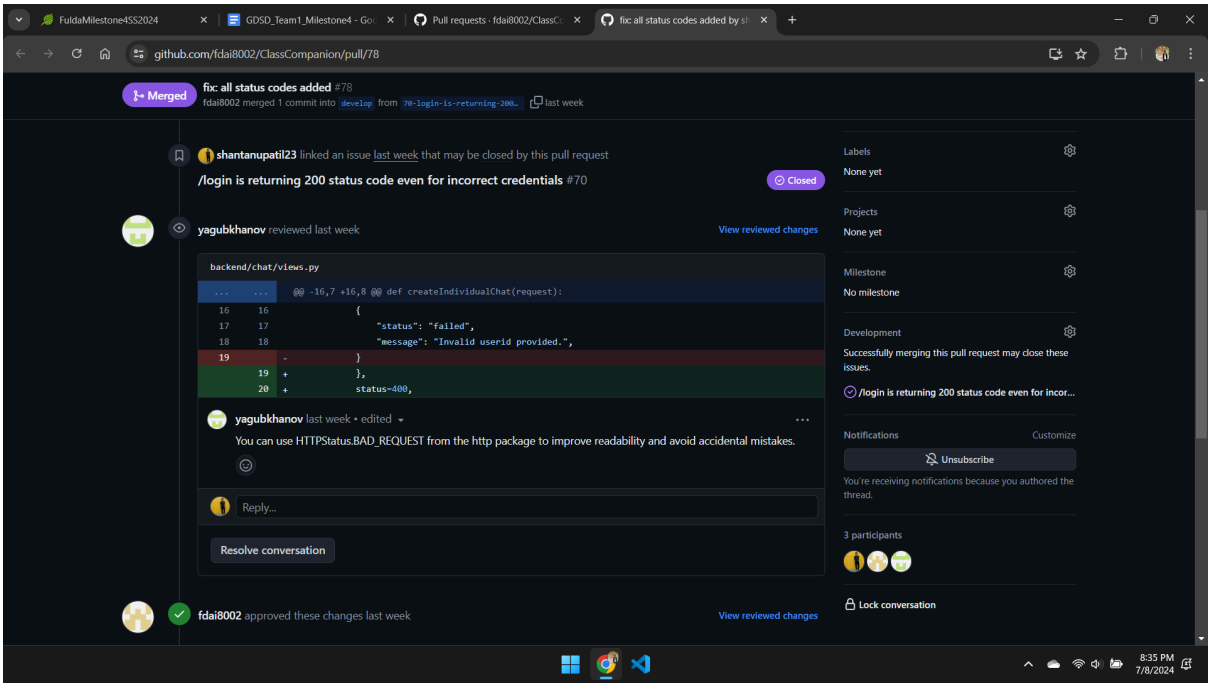
https://github.com/fdai8002/ClassCompanion/pull/63:



https://github.com/fdai8002/ClassCompanion/pull/89:

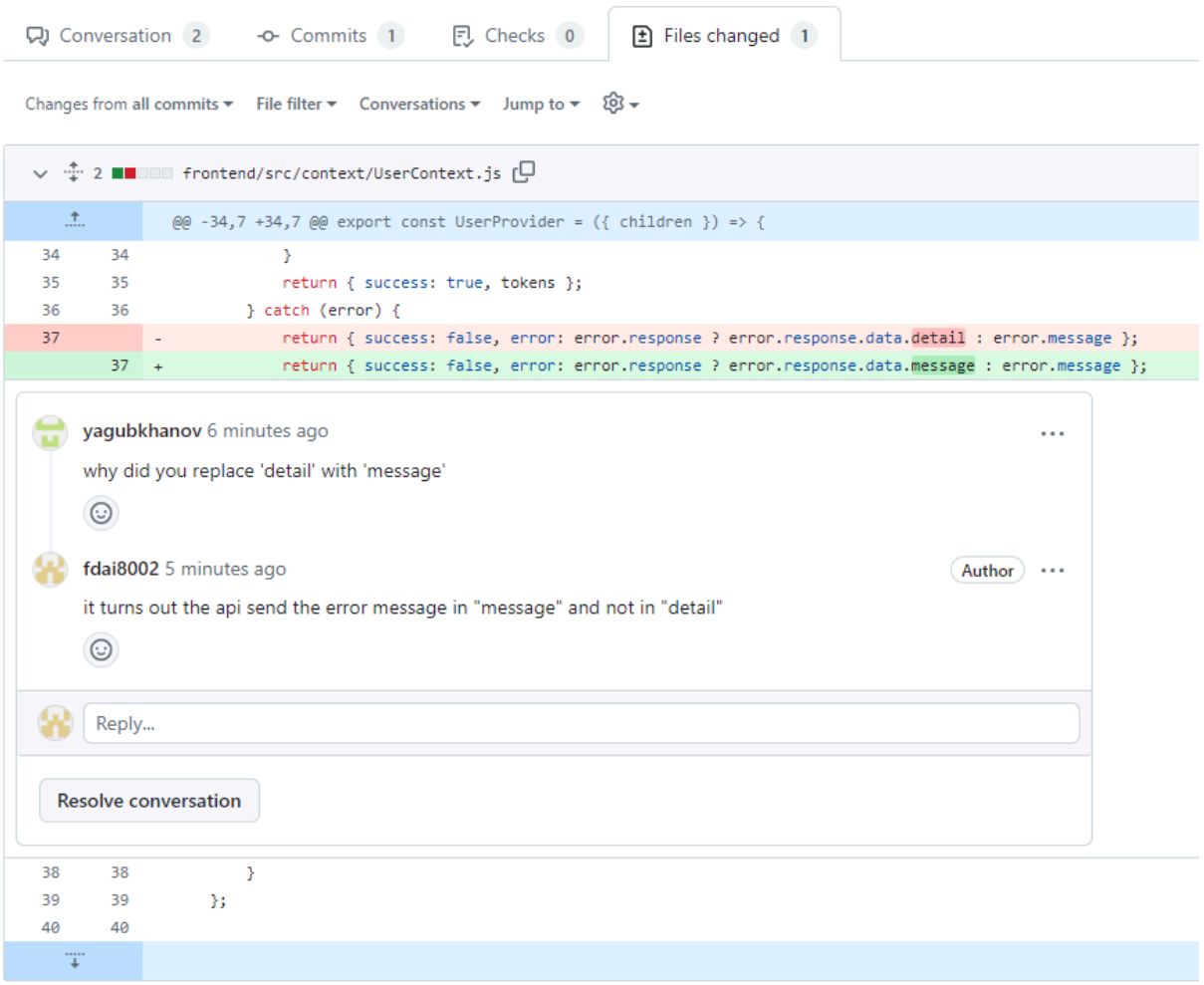


https://github.com/fdai8002/ClassCompanion/pull/78:

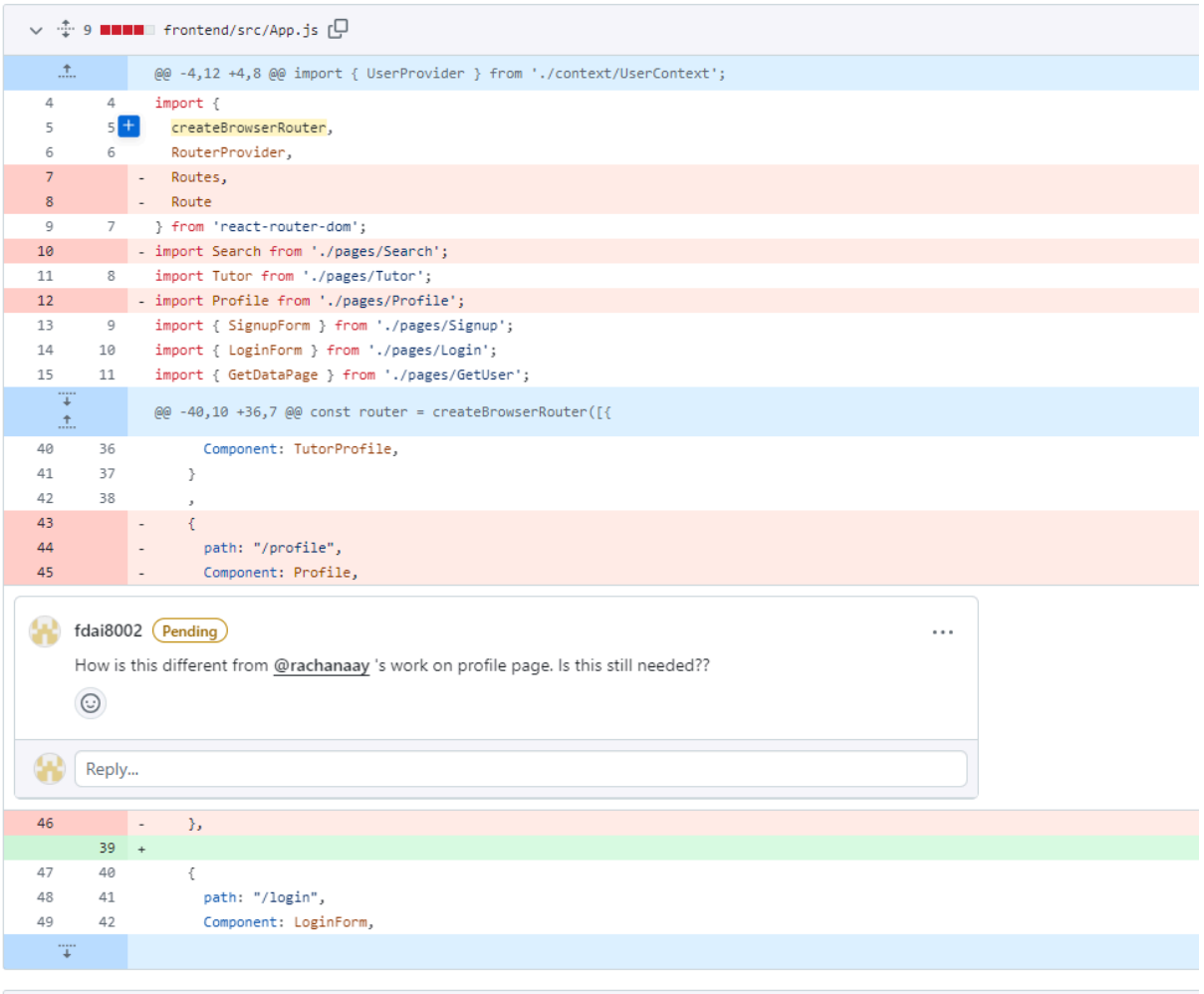


Samman Adhikari (fdai8002) code review:

Samman's PR reviewed by Yagub (https://github.com/fdai8002/ClassCompanion/pull/83)



Amber's PR reviewed by Samman (https://github.com/fdai8002/ClassCompanion/pull/77)



Shantanu’s PR reviewed by Samman (https://github.com/fdai8002/ClassCompanion/pull/84)

d files

nagement

_view.py

backend/chat/views.py

```
@@ -3,6 +3,7 @@
3  3  from rest_framework.decorators import api_view
4  4  from rest_framework_simplejwt.authentication import JWTAuthentication
5  5  from .models import ChatMessage, ChatGroup, User
   6  + from user_management.models import Multimedia
6  7
7  8
8  9  @api_view(["GET"])
@@ -106,8 +107,14 @@ def chatLobby(request):
106 107              chat_type = chat_group.chat_type
107 108
108 109              if chat_type == "personal":
109     -                name = chat_group.users.exclude(id=user.id).first().username
110     -                picture = None
    110 +                sender = chat_group.users.exclude(id=user.id).first()
    111 +                name = sender.username
    112 +                try:
    113 +                    picture = Multimedia.objects.get(
    114 +                        user=sender, media_type="profile_picture"
```

fdai8002 5 minutes ago

what key will this be send in??

fdai8002 1 minute ago

This will cause error because you are returning a database object.

Reply...

Resolve conversation

```
    115 +                )
    116 +            except Exception:
```

Aralikere Yogananda Gunarachana (rachanaay) Code Reviews:

Reviewed Amber Asad's code

rachanaay reviewed 2 minutes ago — View reviewed changes

frontend/src/pages/Tutor.jsx

```
119  +    setError(null); // Clear any previous errors
120  +    try {
121  +      const response = await axios.get(
122  +        `http://localhost:8000/search?query=${searchTerm}`
```

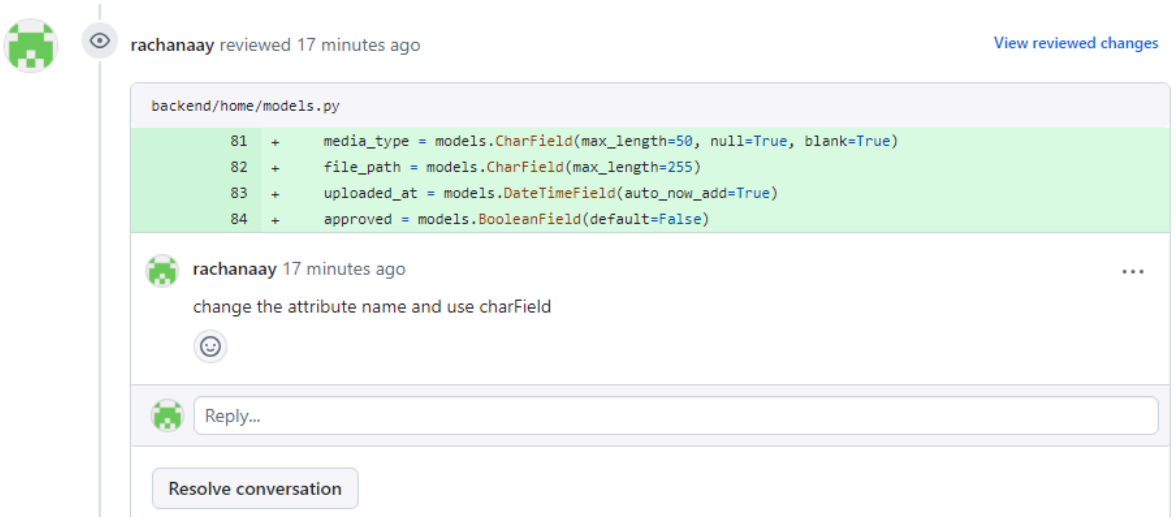
rachanaay 2 minutes ago

Update the Axios request URL to use process.env.REACT_APP_WS_URL

Reply...

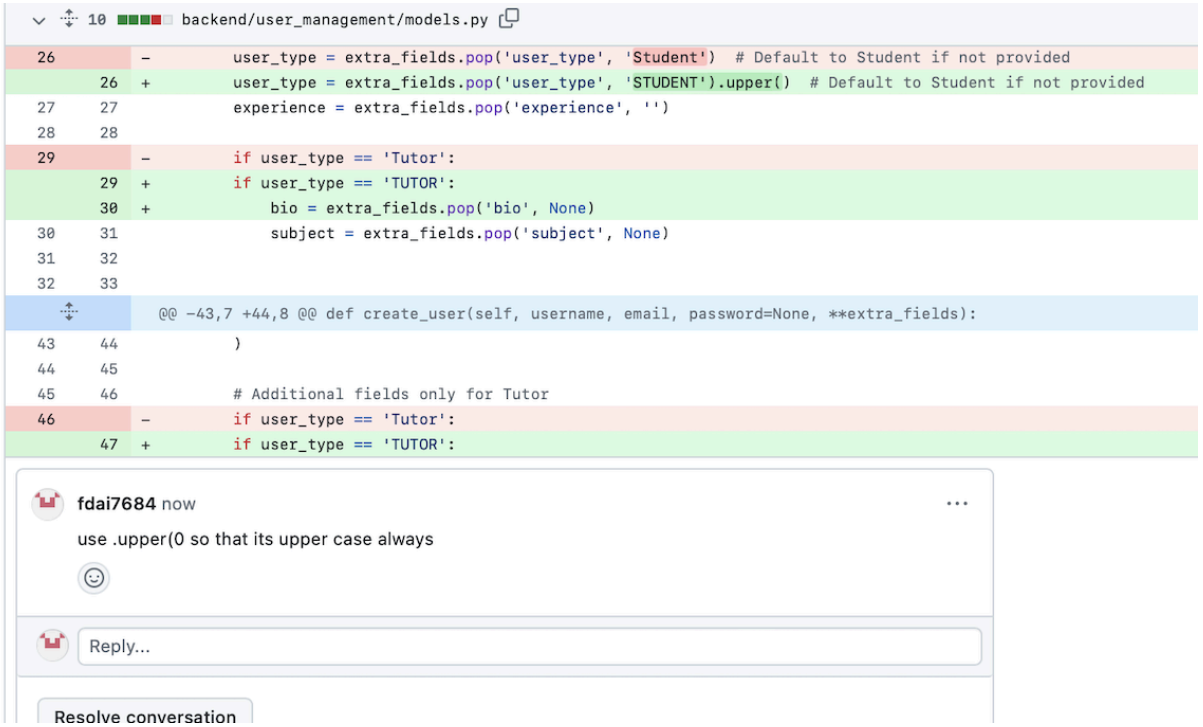
Resolve conversation

Reviewed Yagub's Code



Mariha Asif's (fdai7684) Code Reviews:

Reviewed Shantanu's code



Reviewed Yagub's code



Yagub reviewd Mariha's code



5. Self-check on Best Practices for Security

1. User Authentication and Authorization

• Strong Password Policies: Enforced complex passwords and encourage regular updates.

• Role-Based Access Control (RBAC): Restrict access based on user roles to minimize the risk of unauthorized access.

2. Data Encryption

• Transport Layer Security (TLS): Used HTTPS to encrypt data in transit between clients and servers.

• End-to-End Encryption (E2EE): Implemented E2EE to ensure that only the communicating users can read the messages.

3. Secure Communication Channels

• WebSockets Security: using WebSockets for real-time communication, ensure they are secured with WSS (WebSocket Secure).

4. Server Security

• Regular Software Updates: all server software and dependencies are up to date to mitigate vulnerabilities.

5. Secure Coding Practices

• Input Validation and Sanitization: validated and sanitized user inputs to prevent SQL injection and other code injection attacks.

• Use Secure Libraries and Frameworks: Chose well-maintained and secure libraries and frameworks for development.

• Avoiding Hard-Coding Secrets: Stored secrets like API keys and passwords in secure vaults or environment variables, not in the codebase.

• Server-side validation checks for valid characters and length restrictions (up to 40 alphanumeric characters) in search bar and prevents injection attacks and maintain data integrity.

6. User Privacy and Data Protection

• Data Minimization: Collected only the necessary user data and limit the storage period.

7. Secure User Interface

• Input Filtering: Implemented client-side input filtering to complement server-side validation.

• Escape Outputs: Ensure all user-generated content is properly escaped to prevent XSS.

8. Session Management

• Token-Based Authentication: Use tokens (e.g., JWT) for managing sessions securely.

9. Monitoring and Logging

• Regular Audits: Conducted regular security audits and vulnerability assessments to identify and address weaknesses.

10. Disaster Recovery and Incident Response

• Backup and Restore Procedures: Implemented regular backups and have a robust data recovery plan.

• Incident Response Plan: Developed and maintained a plan to quickly respond to security incidents.

By adhering to these security practices, ClassCompanion ensures robust protection of user credentials, data integrity, and application endpoints against common security threats. Regular updates and security audits further strengthen these measures to maintain a secure environment for users and data.

6. Self-check on Adherence to Original Non-functional Specifications:

1. Application shall be developed, tested and deployed using tools and servers approved by Class CTO and as agreed in Milestone 0. Application delivery shall be from chosen cloud server. **DONE**

2. Application shall be optimized for standard desktop/laptop browsers e.g. must render correctly on the two latest versions of two major browsers. **DONE**

3. All or selected application functions must render well on mobile devices. **DONE**

4. Data shall be stored in the database on the team's deployment cloud server. **DONE**

5. No more than 50 concurrent users shall be accessing the application at any time. **DONE**

6. Privacy of users shall be protected and all privacy policies will be appropriately communicated to the users. **DONE**

7. The language used shall be English (no localization needed). **DONE**

8. Application shall be very easy to use and intuitive. **DONE**

9. Application should follow established architecture patterns. **DONE**

10. Application code and its repository shall be easy to inspect and maintain. **DONE**

11. Google analytics shall be used (optional for Fulda teams). **ISSUE**

12. No e-mail clients shall be allowed. **DONE**

13. Pay functionality, if any (e.g. paying for goods and services) shall not be implemented nor simulated in UI. **DONE**

14. Site security: basic best practices shall be applied (as covered in the class) for main data items. **DONE**

15. Application shall be media rich (images, video etc.). Media formats shall be standard as used in the market today. **DONE**

16. Modern SE processes and practices shall be used as specified in the class, including collaborative and continuous SW development. **DONE**

17. For code development and management, as well as documentation like formal milestones required in the class, each team shall use their own GitHub to be set-up by class instructors and started by each team during Milestone 0. **DONE**

18. The application UI (WWW and mobile) shall prominently display the following exact text on all pages "Fulda University of Applied Sciences Software Engineering Project, Fall 2021 For Demonstration Only" at the top of the WWW page. (Important so as to not confuse this with a real application). **DONE**